Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms. The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U, f)], [I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 (Positive tensor presentation)**.** *Let T_+ and $T_{\mu,+}$ be the sources of Ψ and $\mu\Psi$, respectively. All definitions below use the original ground $\mathbb{S}$. For a* RES *object with underlying set X, put*

$$\Lambda_X = \{\delta \in \Gamma \cap \mathrm{dcl}^{\mathrm{eq}}(\mathbb{S}) : \exists\ \mathbb{S}\text{-definable } c : X \longrightarrow \delta^\sharp\}.$$

Let $\mathcal{C}$ be the semiring congruence on T_+ generated by

$$[\boldsymbol{X}]_r \otimes [U]_m \ \sim\ [\boldsymbol{X}]_r \otimes [U + \delta e_i]_m, \qquad \boldsymbol{X} \in \mathrm{RES}[r],\ U \in \Gamma[m],\ \delta \in \Lambda_X,$$

where $m \geq 1$ and i is a coordinate index. Let $\mathcal{C}_\mu$ be the semiring congruence on $T_{\mu,+}$ generated by

$$[\boldsymbol{X}]_r \otimes [(U,\sigma)]_m \ \sim\ [\boldsymbol{X}]_r \otimes [(U + \delta e_i, \sigma_\delta)]_m, \qquad \sigma_\delta(\eta) = \sigma(\eta - \delta e_i) - \delta,$$

with $\boldsymbol{X} = (X, f, \omega) \in \mu\mathrm{RES}[r]$ and the same admissibility condition $\delta \in \Lambda_X$. The homomorphisms Ψ and $\mu\Psi$ are surjective, with kernel congruences $\mathcal{C}$ and $\mathcal{C}_\mu$, respectively. They induce isomorphisms of graded semirings

$$T_+/\mathcal{C} \ \simeq\ \mathbf{K}_+\,\mathrm{RV}[*], \qquad T_{\mu,+}/\mathcal{C}_\mu \ \simeq\ \mathbf{K}_+\,\mu\mathrm{RV}[*].$$

Proof. The finite product normal form over the original ground partitions every $\mathrm{RV}[k]$ object into pieces isomorphic to $X \times U^\sharp$, where X has finite Gamma support and has a finite label of degree $k - m$, and $U \in \Gamma[m]$. For a measured object the combined weight on a RES factor has finite original-ground Gamma image. Partition by these values and use the shared finite measured Gamma factor in degree zero to transfer each constant weight to the Gamma factor. We may therefore normalize every RES combined weight to 0 and retain the whole weight as a definable function λ on its Gamma factor. This gives surjectivity of both positive tensor maps, including finite and degree-zero factors.

For an admissible δ, choose a witness $c : X \longrightarrow \delta^\sharp$. Multiplication of the ith free RV coordinate by $c(x)$ gives an original-ground definable bijection between the two product lifts; division by $c(x)$ is its inverse. Its Gamma-Jacobian is δ, so the prescribed target form σ_δ makes the map measured. Thus the displayed generating pairs lie in the respective kernels. The pair depends on δ, not on the choice of witness.

For the converse, work in one total degree k and represent the two positive tensors by finite disjoint unions of product objects. Let m be the largest Gamma arity on either side of a definable RV bijection F between the product unions. In total degree k, all maximal-arity factors have RES degree $k - m$. We induct on m.

At a full-rank Gamma type in a source product and a residue type in its RES factor, the angular normal form of the graph gives, on a product neighbourhood,

$$F(x,u) = \big(h(x), c(x)u^M\big), \qquad M \in \mathrm{GL}_m(\mathbb{Z}), \qquad \mathrm{vrv}(c(x)) = \boldsymbol{\delta} \text{ constant}.$$

Here the Gamma rank and the total degree force the target free arity and the RES degree to agree with those of the source. The residue output depends only on x. After fixing x, the generic full-fibre monomial calculation gives coefficients over $\mathbb{S}(x)$, which are retained as definable functions of x. Injectivity forces the square exponent matrix to be unimodular. Finite Gamma support makes the values of those coefficients constant after a residue partition, and their values are definable over the original ground. Whole-fibre surjectivity then forces h to be injective; shrinking and taking its image makes it a RES bijection. Product completeness and compactness extend these formulas to definable product neighbourhoods. No original-ground RV representative of $\boldsymbol{\delta}$ is chosen.

Compactness supplies finitely many such product neighbourhoods, with exceptions in finitely many proper original-ground integral-character hyperplanes. A branch with smaller target Gamma arity lies entirely over such hyperplanes. Apply the same construction to F^{-1} and reverse those matches too. A common finite Boolean partition of the source RES and Gamma factors selects disjoint product cells. The two-sided cut property gives product images and retains every original

patch image, including the inverse cover. On each matched cell, the integral monomial part is already a Gamma morphism and the elementary pairs handle the coefficient translations one coordinate at a time.

Keep the matched target boxes whole and partition their complements by a finite product grid. These are positive tensor identities on both sides. Restricting F to the actual two complements supplies the residual comparison directly. Every nonempty residual maximal-arity Gamma factor is covered by the finite exceptional hyperplanes. Partition by the first applicable hyperplane. A primitive integral equation and a unimodular coordinate change put one coordinate at a fixed original-ground value. Finite-Gamma balancing transfers that coordinate into RES, lowering Gamma arity by one and preserving total degree. The old lower-arity factors already satisfy the smaller bound. Apply induction to the residual tensors and add the matched congruences. For $m = 0$, all Gamma factors are finite and are absorbed into RES, where the positive RES embedding into RV gives equality. Finally reassemble the finitely many total degrees. This proves the unmeasured kernel assertion without cancelling matched summands.

For the measured kernel, use the preceding normalization on both sides. A matched rectangle then satisfies

$$\lambda(\gamma) = \nu(M\gamma + \boldsymbol{\delta}).$$

The unimodular Gamma morphism and the elementary translated pairs respect these complete combined weights. Retain each parent's Gamma weight on every cell of the finite product grid. On the disjoint union of native charts, the forms glue to an original-ground definable form; the complement inherits this form, and the restricted map F preserves it. The two residual positive sums therefore present actual measured RV objects. In the hyperplane reduction, give the fixed finite Gamma prefix combined weight 0 and substitute its value into the remaining Gamma weight function. Balancing transfers its degree into RES while retaining the full remaining weight. At rank zero absorb the actual finite measured Gamma object with its entire form into measured RES. The same positive induction and finite-degree reassembly prove the measured kernel assertion.

Every step uses actual bijections, finite partitions, and positive semiring operations. Arithmetic in the Gamma-valued forms changes object data; it does not subtract semiring classes. The quotient isomorphisms follow from surjectivity and the kernel-congruence equalities, without additive cancellation or groupification. □

T001: why general U is retained at this stage. The ray-only presentation is a later ring-ideal simplification. It is insufficient as a positive congruence presentation. Indeed collapse all unbounded positive tensor classes to one class and retain equality on bounded classes. Boundedness and emptiness give a semiring congruence of this form: the unbounded region absorbs addition and multiplication by nonzero classes, and zero remains separate. All nontrivial ray pairs lie in that region. Nevertheless, over $\mathbb{C}((t))$, the bounded tensors

$$[R]_0 \otimes [(0, \delta]]_1, \qquad [R]_0 \otimes [(\delta, 2\delta]]_1, \quad R = \{r : r^5 = \mathrm{rv}(t)\}, \quad \delta = 1/5,$$

are distinct by the endpoint detector (values -5ε and 5ε) and have equal RV images by multiplication by r. Thus bounded U cannot be discarded. This does not yet prove that every Gamma arity is indispensable.

Formalization status. Both positive kernel equalities and graded quotient semiring isomorphisms are checked in Lean at the explicit canonical, saturated, small Gamma-ground scope; general imaginary-ground transport remains separate. This is the same T001 review item, and the approved working manuscript remains unchanged.

Note that, however, [8, Proposition 10.10(2)] does not hold. One of the main motivations of this paper is to construct an alternative to be used in applications.